

AI-Driven Water Management System for Agriculture Purpose in Sri Lanka

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A.T.R.Jayawardena
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Supervised by

Dr Indrani Kularathne

in collaboration with the Government Factory, Sri Lanka



**Department of Mechanical Engineering
Faculty of Engineering, University of Peradeniya**

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Executive Summary

This project presents the design, development, and implementation of an Artificial Intelligence (AI)-driven water management system tailored to the precision irrigation requirements of paddy agriculture in Sri Lanka's dry zone. Sri Lankan agriculture faces critical challenges including high water duty, the head-tail inequity problem in canal-based irrigation, and erratic climate patterns that undermine traditional, calendar-based water release decisions.

The developed system integrates multiple engineering disciplines — civil and agricultural engineering, machine learning, IoT hardware design, and web application development into a unified decision-support platform. The core components include: (i) a real-time crop water demand calculation engine implementing the FAO-56 Penman-Monteith methodology; (ii) a soil moisture threshold calculator based on soil-physics parameters; (iii) a research framework for LSTM-based soil moisture prediction using meteorological data; (iv) a multi-criteria irrigation decision model based on the Analytic Hierarchy Process (AHP) for collective field management; (v) a custom-designed IoT sensor node utilising an ESP32 microcontroller with soil moisture, temperature, and water level sensors transmitting data to Firebase Cloud; and (vi) a full-stack web application integrating all computational modules with live sensor telemetry and real-time weather API data.

The system provides a five-day irrigation water demand forecast derived from OpenWeatherMap and Open-Meteo environmental APIs, combined with crop-specific coefficients and field soil properties. The ML-based soil moisture prediction and the AHP-based group decision model are developed as research components, reserved for future field-level validation once sufficient real-world data is available. The IoT hardware was successfully deployed in a paddy field, demonstrating sensor calibration, power system design, and cloud telemetry. The web application provides a glassmorphism dark-mode interface with real-time dashboard, rainfall visualisation, soil threshold display, and tabular demand estimation. This project establishes a comprehensive, scientifically grounded proof-of-concept for AI-driven precision irrigation in Sri Lanka.

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Disclaimer

This project report has been prepared as a partial fulfilment of the requirements for the degree of Bachelor of the Science of Engineering at the University of Peradeniya. The work described herein is original and was carried out by the author under the supervision of the named academic supervisors and in collaboration with industrial partners.

The computational models, algorithms, and system architectures developed in this project are intended as proof-of-concept implementations for academic purposes. The machine learning models for soil moisture prediction and the Analytic Hierarchy Process decision framework presented herein are research components that have not yet been validated against full-season real-world field data, and are not intended for immediate commercial deployment without further empirical calibration.

All third-party data, APIs, libraries, and references used in this project have been appropriately cited. The author accepts no liability for any consequences arising from the use of information contained within this report.

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List of Abbreviations

Abbreviation

ADC
AHP
AMS1117
ANFIS
API
ANN
AWD
CI
CR
CWR
DL
DSS
EC
ET₀ / ET₀
ET_c
FAO
FC
FO
GPIO
HC-SR04
HTML
HTTP
HTTPS
IoT
JSON
K_c
LAI
LDO
LM2596
LoRa
LoRaWAN
LSTM
MAD
ML
MPC
NDVI
PCB
PWM
PWP
RAW
RE
RF
RI
RL
RNN
SCMARL

SMP
SPA
SVR
TAW
VWC

Full Form

Analogue-to-Digital Converter
Analytic Hierarchy Process
AMS1117 Low-Dropout Voltage Regulator
Adaptive Neuro-Fuzzy Inference System
Application Programming Interface
Artificial Neural Network
Alternate Wetting and Drying
Consistency Index
Consistency Ratio
Crop Water Requirement
Deep Learning
Decision Support System
Electrical Conductivity
Reference Evapotranspiration
Crop Evapotranspiration
Food and Agriculture Organisation
Field Capacity
Farmer Organisation
General Purpose Input/Output
Ultrasonic Distance Sensor Module
HyperText Markup Language
HyperText Transfer Protocol
HyperText Transfer Protocol Secure
Internet of Things
JavaScript Object Notation
Crop Coefficient
Leaf Area Index
Low-Dropout Voltage Regulator
LM2596 Step-Down DC-DC Converter
Long Range Radio
Long Range Wide Area Network
Long Short-Term Memory
Management Allowable Depletion
Machine Learning
Model Predictive Control
Normalised Difference Vegetation Index
Printed Circuit Board
Pulse Width Modulation
Permanent Wilting Point
Readily Available Water
Effective Rainfall
Random Forest
Random Index
Reinforcement Learning
Recurrent Neural Network
Semi-Centralised Multi-Agent
Reinforcement Learning
Soil Matric Potential
Single Page Application
Support Vector Regression
Total Available Water
Volumetric Water Content

Wi-Fi
XGBoost

Wireless Fidelity
Extreme Gradient Boosting

1. Introduction

Water scarcity and inefficient irrigation management represent two of the most critical challenges facing agricultural sustainability in Sri Lanka. The nation's paddy cultivation sector, which accounts for the majority of inland water withdrawals, is characterised by high water duty and low overall irrigation efficiencies — in some schemes as low as 35% against a sustainability target of at least 65% (Salman et al., 2023). These inefficiencies arise from decisions rooted in traditional calendar-based scheduling rather than precise, data-driven analysis of crop water requirements, soil moisture states, and weather forecasts.

The objective of this project is to design and develop an AI-driven water management system specifically adapted to the structural and agronomic context of Sri Lankan paddy irrigation. The system addresses multiple levels of irrigation decision-making — from daily crop water demand computation to multi-field release scheduling — and integrates physical field monitoring through an IoT sensor network with a web-based decision-support interface.

1.1 Project Aims and Scope

The primary aims of this project are as follows:

- To compute irrigation water demand for paddy fields using the internationally validated FAO-56 Penman-Monteith methodology, integrating predicted meteorological parameters from environmental APIs.
- To research and develop a framework for soil moisture prediction using Long Short-Term Memory (LSTM) deep learning networks trained on meteorological time-series data.
- To design and implement a multi-criteria irrigation scheduling decision model based on the Analytic Hierarchy Process (AHP) for collective field tract management.
- To design, build, and deploy an IoT sensor node based on the ESP32 microcontroller for real-time field monitoring of soil moisture, ambient temperature, and field water level.
- To develop a full-stack web application that integrates real-time IoT telemetry, weather API data, and the crop water demand computation engine into an accessible user interface.

1.2 Intended Beneficiaries

The primary beneficiaries of this system are the Farmer Organisations (FOs) and irrigation officials responsible for managing communal irrigation tracts (locally known as 'Yaya') in Sri Lanka's dry-zone irrigation schemes. Secondary beneficiaries include

individual smallholder paddy farmers who rely on timely and equitable water releases, as well as the Department of Irrigation and agricultural extension officers who support water management planning.

1.3 Scope of the Project

This project encompasses the following components within its defined scope:

- Agricultural and civil engineering knowledge acquisition relating to Sri Lankan paddy irrigation.
- Implementation of crop water demand calculation using FAO-56 equations and API-sourced weather data.
- Research on machine learning models (LSTM) for soil moisture prediction with LSTM model code development.
- Design of an AHP-based multi-criteria decision framework for group field irrigation scheduling.
- Hardware design and construction of an IoT sensor node for field deployment.
- Full-stack web application development integrating all the above components.

The ML soil moisture prediction model and the AHP decision framework are developed as research components and reserved for future integration, as the project recognises the current lack of sufficient real-world field data and the time constraints of a final-year project.

1.4 Approach

The project follows a product-oriented development methodology. An extensive background study of irrigation engineering in Sri Lanka was conducted first, integrating civil hydraulic engineering principles with agro-hydrological physics. This was followed by the design and implementation of the demand calculation engine, the IoT hardware, and the web application. Simultaneously, a research thread investigated ML and AHP frameworks. The system was incrementally tested and verified at each stage, culminating in a field deployment of the IoT sensor node.

2. Background

2.1 Problem Statement

Sri Lankan paddy agriculture faces multiple inter-related problems that collectively result in severe water resource mismanagement. These problems are documented in the context of both individual field operations and system-wide irrigation governance.

2.1.1 High Water Duty and Resource Inefficiency

Traditional paddy cultivation in Sri Lanka is based on continuous inundation of fields, requiring far more water than the actual crop requirement. Average irrigation efficiency in some areas is as low as 35%, against a sustainability target of at least 65% (Salman et al., 2023). Decisions are typically made on the basis of visual analysis of field conditions or fixed seasonal calendars rather than data-driven predictions of crop water needs.

2.1.2 The Head-Tail Inequity Problem

Water distribution in large irrigation schemes is frequently inequitable due to the geographical position of fields along the canal. Farmers at the 'head' of the canal often extract more water than their entitlement, causing severe water deficits for 'tail-end' farmers. This head-tail inequity leads to abandoned paddy fields and reduced productivity in the lower reaches of the irrigation system. An automated, data-driven scheduling system can provide a transparent and equitable release schedule that prevents this over-extraction.

2.1.3 Operational Complexity of the Yaya System

As is characteristic of Sri Lankan irrigation, water is released to an entire tract or 'Yaya' — a communal group of fields — rather than to a single plot. Sri Lankan irrigation systems are characterised by high structural density, making them complex to operate. While individual farmers may have differing water needs, the Farmer Organisation (FO) must make a single collective decision to open the main sluice gate. No current mathematical model exists to aggregate the moisture needs of multiple heterogeneous fields into a single optimal release decision (Chemjong & Wijesekera, 2017).

2.1.4 Information Gaps in Kanna Meetings

Seasonal cultivation decisions in Sri Lanka are made at meetings called 'Kanna meetings'. While legally recognised, these meetings often lack access to accurate real-time data on soil moisture, weather forecasts, or reservoir capacity. Inaccurate forecasting delays water releases compared to actual crop needs. The absence of modern

water-measuring devices at the field level makes it impossible to quantify water use or waste with precision.

2.1.5 Climate Vulnerability

Many irrigation systems in the Dry Zone are under additional pressure from environmental degradation. Reservoirs (tanks) are losing storage capacity due to siltation and encroachment. Simultaneously, rainfall is becoming more erratic, with longer dry spells and higher-intensity flood events. Without precise, ML-based demand forecasting, a single incorrect release decision can deplete an entire tank, leaving the field with no water for the remainder of the season (Walpita, 2021).

2.2 Irrigation Context in Sri Lanka

Sri Lanka's dry-zone paddy cultivation is divided into two primary cultivation seasons:

- **Maha Season (Wet Season, October to March):** Cultivation is timed to capture monsoonal rains. The recommended guideline-based water duty is 1,300 mm; however, actual water issues in major schemes such as Rajangana have been recorded at an average of 2.18 m (Chemjong & Wijesekera, 2017).
- **Yala Season (Dry Season, April to September):** Cultivation relies almost entirely on water stored in reservoirs. The recommended guideline-based water duty is 1,750 mm, with actual field utilisation reaching 2.40 m due to canal conveyance losses and over-irrigation.

The total depth of irrigation water issued to fields over an entire season is defined as the seasonal water duty. A gravity-flow irrigation network consists of hierarchically nested open channels designed to convey, distribute, and apply water across river basins in the sequence: Trans-basin Canal → Main Canal → Distributory Canal → Field Canal → Paddy Plot (Gunawardana, 2025).

Effective rainfall in the dry-zone planning context is estimated using the empirical Irrigation Department formula:

$$RE = 0.67 \times \text{Rainfall}$$

where RE represents the effective rainfall retained in the root zone and contributing to crop growth (Wickramaarachchi, Wijesekera, & Gamage, 2002).

2.2.1 Alternate Wetting and Drying (AWD)

Instead of continuously flooding fields with 5–10 cm of standing water, the Alternate Wetting and Drying (AWD) system allows ponded water to deplete naturally. Irrigation is triggered only when the water table recedes to a critical point of 15 cm below the soil surface. AWD reduces seasonal water consumption by approximately 27%, without impacting grain yield (Wickramaarachchi et al., 2002). This practice forms a key

operational framework supported by the sensor monitoring system developed in this project.

2.3 Soil-Water-Plant Relationships and Soil Physics

Understanding the soil matrix as a three-phase system — composed of solid minerals and organic matter, liquid water, and soil gases — is foundational to agricultural and civil engineering (Gunawardana, 2025). Soil texture refers to the permanent distribution of mineral particles: gravel (>1 mm), sand (1 to 0.05 mm), silt (0.05 to 0.002 mm), and clay (<0.002 mm). Soil structure, unlike texture, is alterable through tillage and directly dictates infiltration rate, root penetration, aeration, and water-holding capacity.

2.3.1 Soil Water Content and Critical Hydrological Thresholds

The quantity of water in a soil profile is expressed using volumetric water content (VWC):

$$De = \theta_v \times \text{Depth of Soil}$$

Crop water availability is bounded by distinct physical soil moisture states along the continuum:

- Field Capacity (FC or θ_{fc}): The maximum water content a soil can retain against gravity after rapid drainage has ceased (1–3 days after saturation). VWC ranges from approximately 20% in sandy soils to 40% in clay soils.
- Permanent Wilting Point (PWP or θ_{wp}): The lower limit of water availability, where soil matrix forces hold water molecules too tightly for plant roots to extract, leading to irreversible wilting. Average soil suction at PWP is 1,500 kPa.
- Total Available Water (TAW): The total water storage capacity of the active root zone: $TAW = (\theta_{fc} - \theta_{wp}) \times z_r$, where z_r is the active rooting depth.
- Management Allowable Depletion (MAD): The fraction of TAW that can be safely depleted before the crop experiences water stress. For highly sensitive paddy rice, MAD is kept at approximately 0.20 (20%) to prevent any yield-limiting water stress (Oklahoma Cooperative Extension Service, 2013).

The soil moisture irrigation trigger threshold is computed as:

$$\theta_{trigger} = FC - [MAD \times (FC - PWP)]$$

This threshold defines the minimum moisture level at which irrigation must be initiated to return the soil to Field Capacity and prevent crop stress.

2.4 Crop Water Requirements and Evapotranspiration

Crop Water Requirements (CWR) represent the depth of water needed to compensate for water losses through evapotranspiration (ET_c) for a healthy crop grown under non-restricting conditions. Evapotranspiration combines two distinct pathways: evaporation

from soil surfaces and standing water, and transpiration — the liquid water extracted by plant roots and released as vapour through leaves.

2.4.1 FAO-56 Reference Evapotranspiration

To standardise crop water calculations, the FAO Penman-Monteith equation is used to estimate reference evapotranspiration (ET_o) from a hypothetical grass reference crop (Allen et al., 1998). Actual crop water requirements are then computed using a crop-specific coefficient (K_c):

$$ET_c = K_c \times ET_o$$

Under stressed field conditions, a water stress coefficient (K_s) is introduced:

$$ET_{c,adj} = K_s \times K_c \times ET_o$$

2.4.2 Biological Water Requirements of Paddy Rice

Paddy rice has highly distinct water demands across its developmental lifecycle, as summarised in Table 2.1 below (Salman et al., 2023; Wickramaarachchi et al., 2002).

Crop Phase	Biological Growth Stage	Water Requirement (mm)	% of Season
Phase 1	Nursery	40 mm	3%
Phase 2	Main Field Preparation	200 mm	16%
Phase 3	Transplanting to Panicle Initiation	460 mm	37%
Phase 4	Panicle Initiation to Flowering	417 mm	34%
Phase 5	Flowering to Maturity	123 mm	10%
Total	Full Life Cycle	1,240 mm	100%

Table 2.1: Paddy Rice Biological Growth Stages and Water Requirements

2.5 Open-Channel Hydraulics and Irrigation Network Efficiency

Due to seepage, evaporation, operational leaks, and channel friction, substantial quantities of water are lost during conveyance. Hydraulic engineers use standardised efficiency metrics to plan system performance (Gunawardana, 2025; Weligamage, 1998):

$$Project\ Efficiency\ (E_p) = E_c \times E_b \times E_a$$

where E_c is the Conveyance Efficiency, E_b is the Field Canal Efficiency, and E_a is the Field Application Efficiency. For major gravity-fed canals in Sri Lanka, the Department of Irrigation recommends an overall conveyance efficiency of 70% ($E_c = 0.70$) (Chemjong & Wijesekera, 2017).

2.6 Machine Learning for Precision Irrigation

Integrating machine learning enables irrigation systems to analyse spatial and temporal weather and soil data to predict water needs proactively (Abioye et al., 2022). Several ML paradigms are relevant to precision irrigation:

2.6.1 Supervised Learning Models

- Multiple Linear and Polynomial Regression: Used to predict daily irrigation water requirements from temperature, rain, and moisture sensor data.
- XGBoost Regressor: Proven to be the most accurate algorithm for meteorological and precipitation forecasting. In localised evaluations, XGBoost achieved 92% prediction accuracy (RMSE $\approx$ 8.53 mm/day) (Parikh & Sharma, 2024).
- Support Vector Regression (SVR): Applied to model non-linear soil moisture content curves by mapping high-dimensional input spaces into optimal linear boundary hyperplanes.
- Decision Trees (DT) and Random Forest (RF): Highly effective for binary classification tasks such as 'irrigate / do not irrigate' decisions.

2.6.2 LSTM Networks for Time-Series Soil Moisture Prediction

Soil moisture behaves as a continuous, smooth time series influenced by historical lag states, weather dynamics, and soil physics. Long Short-Term Memory (LSTM) networks resolve the vanishing gradient problem of standard Recurrent Neural Networks (RNNs) by incorporating specialised memory cell states controlled by three gating mechanisms (Jiang, Niu, & Li, 2024):

- Forget Gate: Decides what proportion of past soil moisture and weather states to discard.
- Input Gate: Identifies which new meteorological features (humidity, wind speed, air temperature, solar radiation) will update the memory cell.
- Output Gate: Generates the predicted volumetric water content (VWC) or soil matric potential (SMP).

An important theoretical finding from the literature (Jiang et al., 2024) is the -40 cm depth cutoff theory:

- Shallow Soil Zone (0 to -40 cm): Highly dynamic and sensitive to evapotranspiration and immediate rainfall infiltration. The LSTM must use a prior sequence model (e.g., the past 60 hourly data points) weighting temperature, wind, and solar radiation.
- Deep Soil Zone (Below -40 cm): Less correlated with short-term weather events. Models can reduce the feature dimension, relying on multidimensional linear fitting with adjacent layers, incorporating non-time-varying parameters like soil bulk density and texture.

2.6.3 Reinforcement Learning for Decision-Making

While supervised learning is excellent for forecasting moisture states, Reinforcement Learning (RL) is used to optimise the actual decision-making policy for water actuators. A Semi-Centralised Multi-Agent RL (SCMARL) framework addresses the scalability challenges of large, spatially variable fields with a two-tier hierarchy (Agyeman, Liu, & Shah, 2024):

- Top-Tier Coordinator Agent: Makes a daily binary decision ($c \in \{0,1\}$) on whether irrigation should occur across the field, based on concatenated soil moisture profiles and global weather forecasts.
- Bottom-Tier Local Agents: One per Management Zone, determining the precise continuous irrigation water depth required for their zone.

2.7 Analytic Hierarchy Process in Multi-Criteria Decision Making

The Analytic Hierarchy Process (AHP) is a structured multi-criteria decision-making methodology developed by Saaty. It decomposes a complex decision problem into a hierarchy of criteria, sub-criteria, and alternatives, using pairwise comparisons to derive priority weights (Forman & Peniwati, 1998).

In the context of irrigation management for a communal paddy tract (Yaya), the AHP is particularly valuable because the water release decision must aggregate the differing needs of multiple fields with varying crop types, crop stages, soil properties, and current moisture levels into a single, mathematically justified collective decision.

3. Specification and Design

3.1 System Overview and Requirements

The developed system is a product-oriented, multi-module platform for AI-driven irrigation management in Sri Lankan paddy agriculture. The overall system architecture integrates four principal engineering layers:

- Physical IoT Sensing Layer: Field-deployed ESP32 sensor node measuring soil moisture, ambient temperature, and field water level.
- Cloud Data Layer: Firebase Realtime Database storing live sensor telemetry and demand calculation logs.
- Computation Layer: Python backend server implementing FAO-56 Penman-Monteith demand calculations, soil threshold analysis, and integration with environmental weather APIs.
- Presentation Layer: A web-based single-page application providing the user interface, real-time dashboards, and irrigation recommendations.

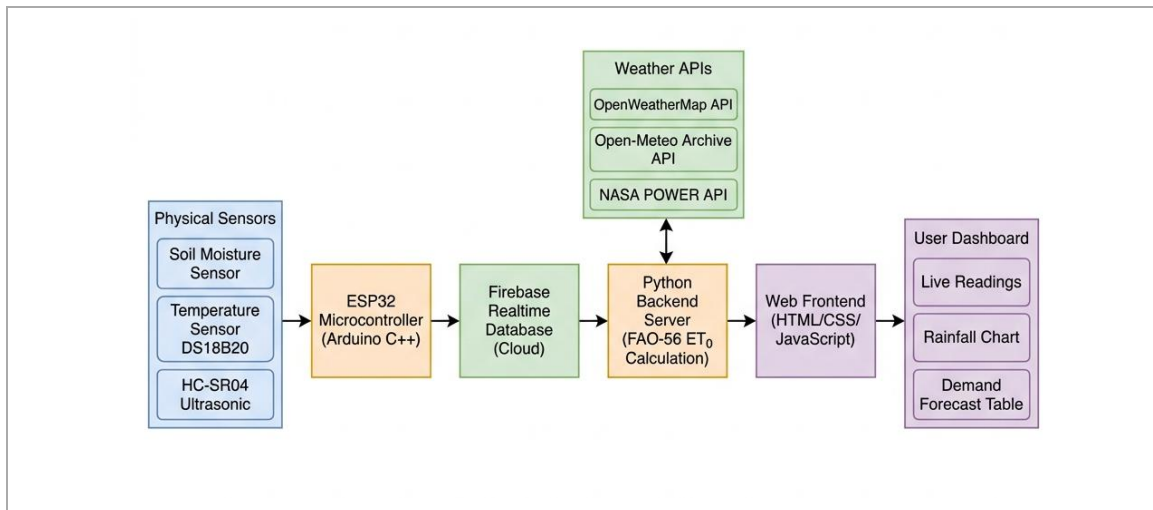


Figure 3.1: Overall System Architecture of the AI-Driven Irrigation Management System

3.1.1 User Requirements

The system must satisfy the following user requirements:

- Provide a five-day irrigation water demand forecast based on real-time weather data and crop parameters.
- Display live soil moisture, temperature, and water level readings from the IoT sensor node.
- Allow configuration of GPS location, soil texture, crop type, field area, channel efficiency, and planting date.
- Calculate and display soil moisture thresholds (FC, PWP, Moisture Threshold) for the specified soil type.
- Store sensor readings and demand calculation logs with timestamps in cloud storage.

- Be accessible via a standard web browser on desktop and mobile devices.
- Operate without dependency on continuous internet connectivity for the IoT sensor node.

3.2 Irrigation Demand Calculation Module

The demand calculation module constitutes the scientifically validated computational core of the system. It implements the FAO-56 Penman-Monteith standard to estimate daily crop water requirements from predicted meteorological parameters obtained from external weather APIs, specifically OpenWeatherMap and Open-Meteo. The environmental parameters available from these APIs are summarised in Table 3.1.

Parameter	API JSON Field	Unit	Available
Temperature	main.temp (OpenWeatherMap)	Kelvin $\rightarrow$ $^{\circ}\text{C}$	Yes
Relative Humidity	main.humidity	%	Yes
Wind Speed	wind.speed	m/s	Yes
Rainfall	rain['3h'] (OpenWeatherMap)	mm	Yes
Cloud Cover	clouds.all	%	Yes
Atmospheric Pressure	main.pressure	hPa	Yes

Table 3.1: Environmental Parameters Available from Weather APIs

NASA POWER API is additionally used for solar radiation data, which is required for accurate net radiation (R_n) computation. Since API data is available at 3-hourly resolution but the FAO-56 equations are daily-based, the computation pipeline aggregates temperature to daily minimum and maximum values and sums radiation to daily totals.

3.2.1 Reference Evapotranspiration (ET_0) Calculation

The FAO-56 Penman-Monteith equation is implemented in the Python backend (demand_calculator.py) to compute the reference evapotranspiration (Allen et al., 1998). The key intermediate computations include:

- Penman-Monteith equation

$$ET_0 = W R_n + (1 - W)f(u_2)(e_s - e_a)$$

- Weighting Factor (W):

$$W = \frac{\Delta}{(\Delta + \gamma)}$$

- Wind Function ($f(u_2)$):

$$f(u_2) = 0.27 \left(1 + \frac{u_2}{100} \right)$$

- Saturation Vapour Pressure (e_s):

$$e_s = 0.6108 \exp \left(\frac{17.27 T}{T + 237.3} \right)$$

- Actual Vapour Pressure (e_a):

$$e_a = (RH/100) e_s$$

- Slope of Saturation Vapour Pressure Curve (Δ):

$$\Delta = \frac{4098 e_s}{(T + 237.3)^2}$$

- Net Shortwave Radiation (R_{ns}): where $\alpha = 0.23$ (albedo)

$$R_{ns} = (1 - \alpha) R_s$$

- Net Longwave Radiation (R_{nl}):

$$R_{nl} = \sigma T_K^4 (0.34 - 0.14 \sqrt{e_a}) \left(1.35 \left(\frac{R_s}{R_{so}} \right) - 0.35 \right)$$

- Net Radiation (R_n):

$$R_n = R_{ns} - R_{nl}$$

When solar radiation data is unavailable directly from the API, the Hargreaves temperature-difference approximation is applied using the empirical constant $k_{Rs} = 0.16$ (Allen, 1997):

$$R_s = k_{Rs} \sqrt{T_{\max} - T_{\min}} R_a$$

where R_a is the extraterrestrial radiation, calculated from solar declination, sunset hour angle, and the inverse square of the Earth-Sun distance using FAO-56 Chapter 3 formulas (Allen et al., 1998).

- Extraterrestrial Radiation (R_a)

$$R_a = \frac{24 \times 60}{\pi} G_{sc} d_r [\omega_s \sin \phi \sin \delta + \cos \phi \cos \delta \sin \omega_s]$$

- Inverse Relative Distance Earth–Sun (d_r)

$$d_r = 1 + 0.033 \cos\left(\frac{2\pi J}{365}\right)$$

- Solar Declination : δ

$$\delta = 0.409 \sin\left(\frac{2\pi J}{365} - 1.39\right)$$

- Sunset Hour Angle: ω_s

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta)$$

3.2.2 Crop Water Requirement and Net Irrigation Demand

The crop coefficient (K_c) is applied as a step-curve function of days after planting, derived from FAO-56 and adapted to Sri Lankan paddy conditions (Lee & Huang, 2014):

$$ET_c = K_c \times ET_o$$

The net irrigation demand accounts for effective rainfall and conveyance losses:

$$\text{Net Irrigation Demand} = (ET_c - RE) / CE$$

where $RE = 0.67 \times \text{Rainfall}$ is the effective rainfall (Wickramaarachchi et al., 2002) and $CE = 0.70$ is the standard dry-zone canal conveyance efficiency (Chemjong & Wijesekera, 2017). This demand is calculated iteratively for each of the five forecast days, producing a tabular output that guides irrigation scheduling decisions.

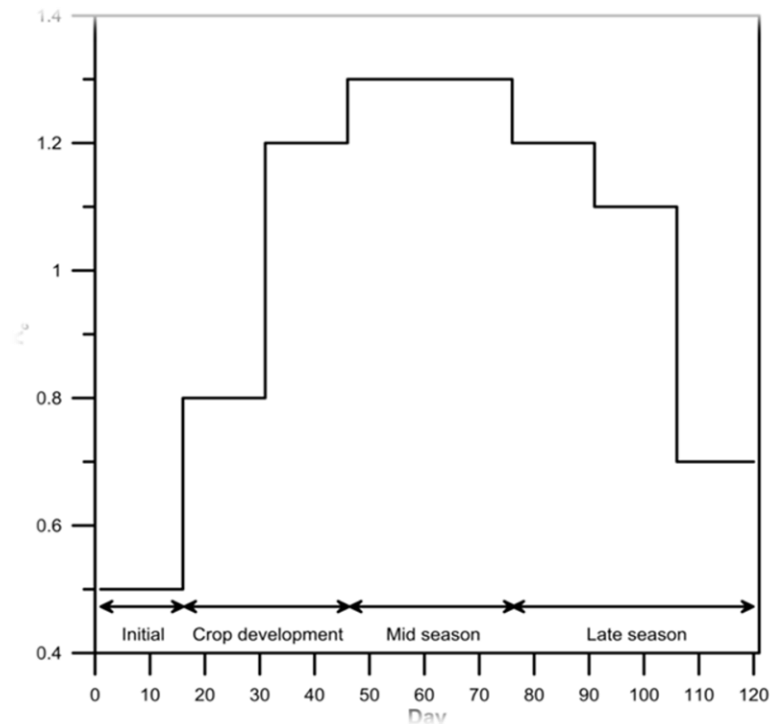


Figure 3.2: Crop Coefficient (K_c) Curve for Paddy Rice by Growth Stage

3.3 Soil Moisture Threshold Calculation

The soil moisture threshold module computes the critical volumetric water content level at which irrigation must be triggered to prevent crop stress. The module requires two primary soil parameters, which are input by the user based on soil texture type:

- Field Capacity (FC): Estimated from soil texture tables. For loam soils: approximately 25–27%; for clay soils: approximately 32–40%.
- Permanent Wilting Point (PWP): Estimated from texture tables. For loam soils: approximately 10–12%; for clay soils: approximately 18–24%. Scientifically, PWP is defined as the state at which soil suction reaches 1,500 kPa (Oklahoma Cooperative Extension Service, 2013).
- Management Allowable Depletion (MAD): For paddy rice, MAD = 0.20 (20%), ensuring that only 20% of the available water is depleted before re-irrigation is triggered.

The trigger threshold is computed as:

$$\theta_{trigger} = FC - [MAD \times (FC - PWP)]$$

This threshold is displayed on the web application's Field Properties screen as part of a visual Soil Moisture Spectrum bar, colour-coded into three zones: Wilting Zone, Depletion Zone, and Optimal Zone (Walpita, 2021; Agyeman et al., 2024).

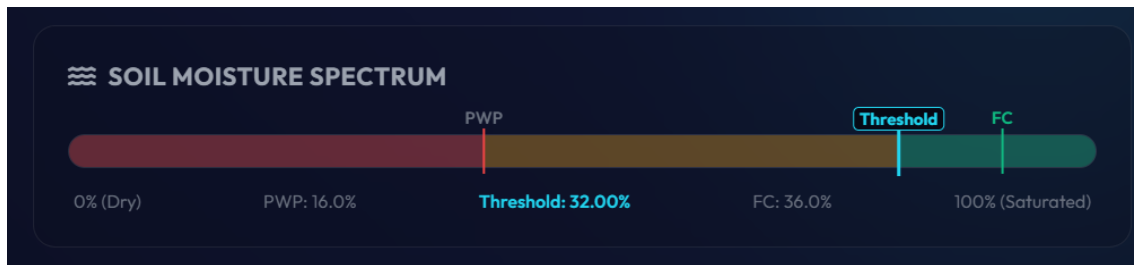


Figure 3.3: Soil Moisture Threshold Diagram

3.4 Machine Learning Research: Soil Moisture Prediction

This section presents the research conducted into machine learning approaches for soil moisture prediction as part of this project. As noted, this component is a research contribution and the developed model code has not been integrated into the deployed system due to the absence of sufficient real-world local field data. However, the theoretical framework and trained model code are documented here as a foundation for future integration.

3.4.1 Data Sources

The machine learning soil moisture prediction model requires two categories of input data:

- Meteorological Data: humidity (%), atmospheric pressure (hPa), wind speed (m/s), wind direction (°), solar radiation intensity (W/m²), accumulated solar radiation (MJ/m²), rainfall (mm), and air temperature (°C).
- Soil Moisture Labels: Volumetric Water Content (VWC) readings, used as the prediction target. Public datasets referenced during research include Kaggle soil moisture time-series datasets.

3.4.2 LSTM Architecture Design

An LSTM network was selected as the primary architecture for time-series soil moisture prediction, based on its demonstrated superiority over standard RNNs for long-range temporal dependencies in soil water dynamics (Jiang et al., 2024). The architecture incorporates the -40 cm depth cutoff principle:

- For the Shallow Soil Zone (0 to -40 cm): A sequence model with a time window of the previous 60 hourly data points is used, weighting temperature, wind speed, and solar radiation.
- For the Deep Soil Zone (Below -40 cm): A reduced feature dimension model incorporating non-time-varying parameters such as soil bulk density and texture ratios.

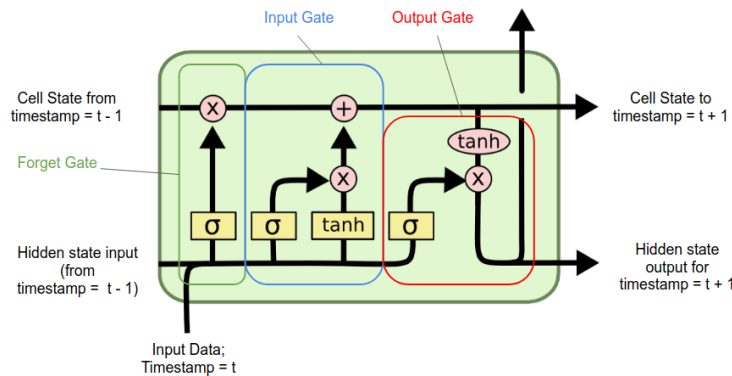


Figure 3.4: LSTM Network Architecture for Soil Moisture Prediction

3.4.3 Reinforcement Learning and SCMARL Framework

Beyond prediction, the research also explored the Semi-Centralised Multi-Agent Reinforcement Learning (SCMARL) framework for irrigation decision optimisation (Agyeman, Liu, & Shah, 2024). This framework employs:

- **Top-Tier Coordinator Agent:** Evaluates field-wide soil moisture conditions and weather forecasts to make a daily binary irrigation decision ($c \in \{0,1\}$), penalising the operational cost of initiating a canal water release.
- **Bottom-Tier Local Agents:** One per Management Zone (MZ), each determining the precise continuous irrigation water depth required for its zone, with its state augmented by the coordinator's binary flag (c) to prevent environment non-stationarity.

State augmentation resolves the non-stationarity challenge of multi-agent learning: the coordinator's binary decision (c) is communicated directly to local agents, forcing them to learn to output zero depth when $c = 0$ and optimise positive depth when $c = 1$. The combined SCMARL-MPC approach converts the complex Mixed-Integer Non-Linear Programming (MINLP) irrigation control problem into a tractable continuous control problem (Agyeman et al., 2024).

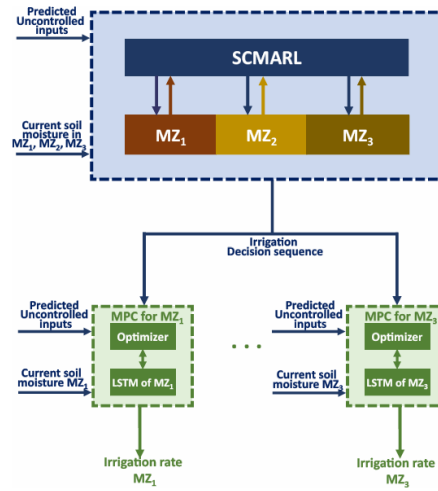


Figure 3.5: SCMARL Framework — Coordinator and Local Agent Hierarchy

3.5 AHP-Based Multi-Criteria Decision Model for Group Field Irrigation

In Sri Lankan paddy irrigation, water is released to an entire Yaya (communal tract) at once, rather than to individual plots. The fundamental challenge is therefore to aggregate the differing irrigation needs of multiple heterogeneous fields — each with different crop types, crop growth stages, soil properties, and current moisture levels — into a single, justified collective decision. The Analytic Hierarchy Process (AHP) was selected to provide this mathematical aggregation framework (Forman & Peniwati, 1998).

3.5.1 Relevant Irrigation Parameters

Based on literature review and consultation with resource persons, eight key parameters were identified as relevant to the irrigation decision:

Parameter	Description	Standard Source	Application
P1	Current Surface Water Level	IRRI AWD Manual	Tracks ponding depth; triggers re-flooding at –15 cm
P2	Root-Zone Soil Moisture (VWC)	FAO Paper No. 56 & IRRI AWD	Measures VWC during dry AWD phases
P3	Water Demand (ETc)	FAO Irrigation & Drainage Paper No. 56	Penman-Monteith daily water demand
P4	Wind Speed	FAO-56 Chapter 3	Drives aerodynamic resistance in ET_o
P5	Deep Drain / Percolation Rate	Bouman et al. (IRRI)	Seepage through puddled plow pan (40–70% of loss)
P6	Ground Water Depth	IRRI Research	Sub-surface water table and capillary rise
P7	Water Quality / Salinity (EC)	FAO Paper No. 29	Electrical conductivity — osmotic stress during flowering
P8	Soil Temperature	Agronomy Literature	Root-zone temperature effects on nutrient uptake

Table 3.2 (Decision Parameters): Key Parameters and Literature Sources for AHP

3.5.2 Stage-Wise Parameter Importance

A critical insight from the research is that parameter importance is not fixed , it varies substantially across the four paddy growth stages. The stage-wise parameter impact example matrix is presented below:

Crop Stage	Primary 80% Vital Parameters	Secondary 20% Parameters
Land Preparation & Seedling (0–15 Days)	P1 (Water Level), P5 (Percolation)	P2 (Soil Moisture), P7 (Salinity)
Vegetative / Tillering (15–45 Days)	P2 (Soil Moisture), P3 (ETc), P1 (Water Level)	P8 (Soil Temp), P7 (Salinity)
Reproductive / Flowering (45–85 Days) — CRITICAL	P1 (Water Level), P7 (Salinity / Quality), P3 (ETc)	P2 (Soil Moisture), P6 (Groundwater)
Ripening / Harvest (85–120 Days)	P1 (Water Level — drainage to 0 cm)	P2, P3, P7 — all near-zero importance

Table 3.3: Stage-Wise Parameter Impact Matrix

3.5.3 AHP Decision Hierarchy

The AHP model is structured as a three-level hierarchy:

- Level 1 — Goal: Select the optimal water release decision for the Yaya on a given date.
- Level 2 — Criteria: The eight field-level parameters (P1–P8), weighted by overall importance derived from expert surveys.
- Level 3 — Alternatives: Each individual field within the Yaya, evaluated against the criteria to compute a composite priority score.

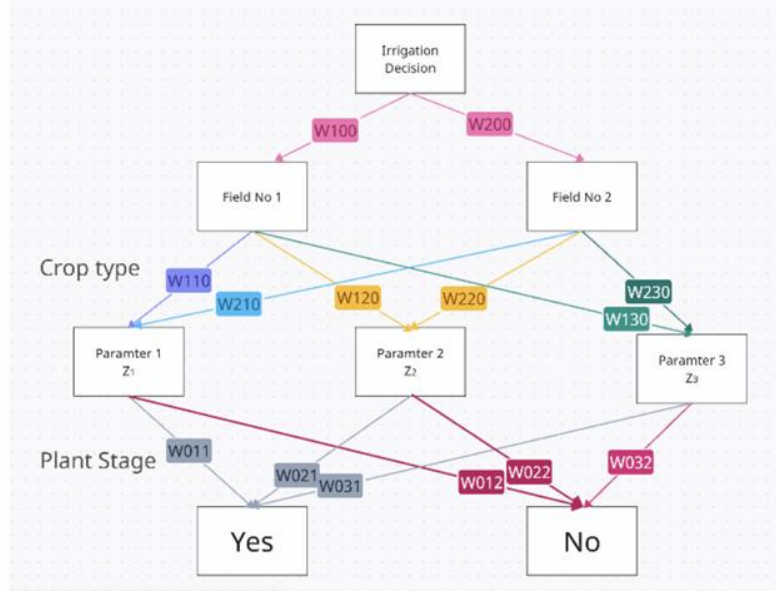


Figure 3.6: AHP Decision Hierarchy for Multi-Field Irrigation Scheduling

3.5.4 Pairwise Comparison and Weight Derivation

For the dominant parameters identified at each crop stage, a pairwise comparison matrix A is constructed on Saaty's 1–9 scale, where a_{ij} represents the relative importance of parameter P_i over P_j . The matrix has the reciprocal property: $a_{ij} \times a_{ji} = 1$. In this report, the parameters are referred to as P1, P2, P3, etc. as defined in Table 3.x, and a_{ij} is the comparison ratio of P_i to P_j (e.g., a_{12} represents how much more important P1 Water Level is compared to P2 Soil Moisture in the current growth stage).

Two distinct comparison matrix sets are constructed by experts:

- Set A (Level 2 Weights): Crop-type-specific parameter comparisons for each field.
- Set B (Level 3 Weights): Plant-stage-specific parameter comparisons for the current growth stage.

Step 3: Column Normalisation

Normalise each column of the pairwise comparison matrix by dividing each element by the column sum:

$$\bar{a}_{ij} = \frac{a_{ij}}{\sum_{i=1}^n a_{ij}}$$

where the denominator is the sum of all entries in column j.

Step 4: Priority Weight Vector (Row Average)

The priority weight w_i of each parameter P_i is the mean of its normalised row:

$$w_i = \frac{1}{n} \sum_{j=1}^n \bar{a}_{ij}$$

The resulting weight vector $w = [w_1, w_2, \dots, w_n]$ gives the relative importance of each parameter, and $\sum w_i = 1.0$. This is the Column Normalisation and Row Averaging (Synthesising) Method (Forman & Peniwati, 1998).

Step 5: Consistency Check

Compute the weighted sum vector (Aw):

$$(Aw)_i = \sum_{j=1}^n a_{ij} w_j$$

Compute the principal eigenvalue $\lambda_{\max}$:

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^n \frac{(Aw)_i}{w_i}$$

Compute the Consistency Index (CI):

$$CI = \frac{\lambda_{\max} - n}{n - 1}$$

Compute the Consistency Ratio (CR) using Saaty's Random Index (RI):

$$CR = \frac{CI}{RI} \quad (\text{accept if } CR \leq 0.10)$$

If $CR \leq 0.10$, the expert judgement is considered acceptably consistent. Responses with $CR > 0.10$ are rejected and the expert is asked to revise their comparisons.

Step 6: Aggregating Filtered Weights Across Multiple Experts

Once individual expert pairwise comparison matrices are validated through the consistency check ($CR < 0.10$), individual priority vectors are synthesized using the Aggregation of Individual Priorities (AIP) method via the Geometric Mean.

1. Calculate Unnormalized Group Geometric Weights (W_i)

For each criterion or parameter i across M validated experts:

$$\bar{W}_i = \left(\prod_{m=1}^M W_i^{(m)} \right)^{\frac{1}{M}} = \sqrt[M]{W_i^{(1)} \times W_i^{(2)} \times \dots \times W_i^{(M)}}$$

2. Normalize Group Weights

To ensure the sum of all final criteria weights equals,

$$\sum_{i=1}^n W_{i,\text{final}} = 1$$

the geometric weights are normalized:

$$W_{i,\text{final}} = \frac{\bar{W}_i}{\sum_{k=1}^n \bar{W}_k}$$

Step 7: Final Decision Logic & Irrigation Trigger Rule

Once the aggregated final criteria weights (W_i) are established via Group AHP and all real-time field parameters are normalized into Water Deficit Indices ($0 < Z < 1$ where 0 represents optimal moisture and 1 represents maximum water stress), the final irrigation trigger state is evaluated.

1. Calculate Field Deficit Index (F_{yes}) and Reference Threshold ($F_{\text{reference}}$)

$$F_{\text{yes}} = \sum_{i=1}^n \omega_i Z_i \quad \text{and} \quad F_{\text{reference}} = \sum_{i=1}^n \omega_i Z_{i,\text{ref}}$$

2. Irrigation Decision Rule

$$\text{Decision Rule} = \begin{cases} \text{Trigger Irrigation (YES),} & \text{if } F_{\text{yes}} > F_{\text{reference}} \\ \text{Hold / Do Not Irrigate (NO),} & \text{if } F_{\text{yes}} \leq F_{\text{reference}} \end{cases}$$

Z_i : Real-time normalized water deficit index for parameter i ($Z_i \in [0, 1]$).

$Z_{i,\text{ref}}$: Agronomic baseline / critical threshold deficit index for parameter i .

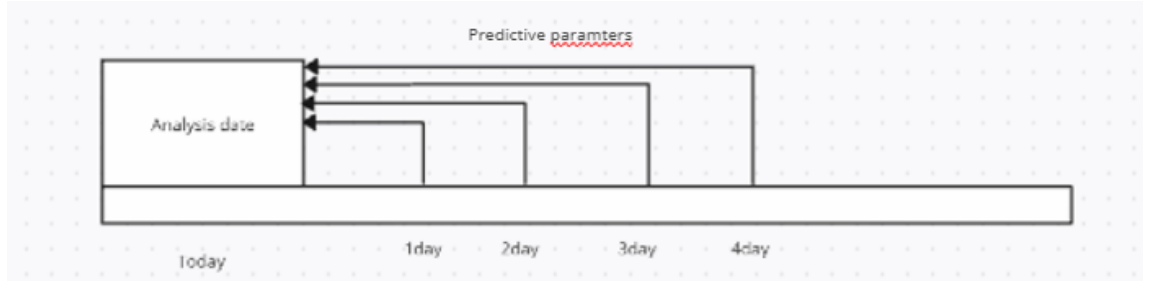


Figure 3.7: Multi-Day Rolling Horizon Framework for Predictive Parameter Ingestion

At the current analysis date (Today), the system implements a rolling multi-day horizon by ingesting forecasted environmental and weather telematics for upcoming days ($t+1$ to $t+4$). Rather than relying solely on instantaneous sensor measurements, the decision framework projects cumulative water deficit trends and upcoming precipitation events across this lookahead window. This enables proactive irrigation scheduling preemptively delivering water before root-zone moisture drops to critical stress thresholds, or withholding delivery if sufficient rainfall is forecasted thereby preventing crop stress while maximizing water and energy savings.

3.6 IoT Hardware Design

The IoT sensor node was designed to provide real-time environmental monitoring from a paddy field and transmit data to the cloud-based Firebase Realtime Database via Wi-Fi. The design prioritised accuracy, power stability, and field resilience.

3.6.1 Core Components

Component	Function	Max Current	Power Source
ESP32 DevKit V1	Primary microcontroller, Wi-Fi	≥ 500 mA (Wi-Fi spikes)	LM2596 5V rail
HC-SR04 Ultrasonic	Water level measurement	20 mA max	LM2596 5V rail
HW-080 / HW-103 Capacitive	Soil moisture (analog)	20 mA max	AMS1117 3.3V rail
DS18B20 OneWire	Ambient temperature	11.5 mA max	AMS1117 3.3V rail
LM2596 Buck Converter	Step-down 7.4V $\rightarrow$ 5V	3 A rated capacity	Battery (7.4V series)
AMS1117-3.3 LDO	Step-down to 3.3V	800 mA max	Battery direct
2 $\times$ 3.7V Li-ion 5800 mAh	Power supply (series $\rightarrow$ 7.4V)	—	—

Table 3.4: IoT Sensor Node Component Summary and Maximum Current Specifications

3.6.2 Power System Architecture

A critical design decision was the transition from a parallel-battery boost converter topology to a series-battery buck converter architecture. Two 3.7 V lithium-ion batteries are connected in series to produce a native supply voltage of 7.4 V, which is then stepped down using a highly efficient (~90%) LM2596 DC-DC buck converter to produce a stable 5 V rail (capable of delivering up to 3 A). This eliminates the brownout resets that plagued earlier designs when the ESP32's Wi-Fi transmission caused 500 mA current spikes.

A dedicated AMS1117-3.3 Low-Dropout (LDO) regulator is branched directly off the battery to power the 3.3 V sensors (DS18B20 and soil moisture). The AMS1117 has a dropout voltage of approximately 1.1 V, requiring a minimum input of 4.4 V, which is satisfied by the series battery configuration (minimum charge $\approx 4.2 \text{ V} \times 2 = 8.4 \text{ V}$; discharged minimum $\approx 3.0 \text{ V} \times 2 = 6.0 \text{ V}$).

Peak system current analysis:

$$I_{Peak} = I_{ESP32} + I_{Ultrasonic} + I_{Soil} + I_{Temp}$$
$$I_{Peak} = 500 \text{ mA} + 20 \text{ mA} + 20 \text{ mA} + 11.5 \text{ mA} = 551.5 \text{ mA}$$

The current distribution:

$$I_{Buck} = 500 \text{ mA} + 20 \text{ mA} = 520 \text{ mA} < 3,000 \text{ mA (LM2596 capacity)}$$
$$I_{LDO} = 20 \text{ mA} + 11.5 \text{ mA} = 31.5 \text{ mA} < 800 \text{ mA (AMS1117 capacity)}$$

3.6.3 HC-SR04 Logic Level Shifting

The HC-SR04 ultrasonic sensor operates at 5 V, producing a 5 V ECHO output signal. Feeding this directly into the ESP32's 3.3 V GPIO pins would cause permanent damage. A resistive voltage divider was designed using three 10 k Ω resistors:

- $R_{top} (R8) = 10 \text{ k}\Omega$
- $R_{bottom} (R9 + R10 \text{ in series}) = 20 \text{ k}\Omega$

$$V_{out} = 5 \text{ V} \times (20 \text{ k}\Omega / (10 \text{ k}\Omega + 20 \text{ k}\Omega)) = 5 \text{ V} \times (2/3) \approx 3.33 \text{ V}$$

This ensures the ECHO signal is safely within the ESP32's 3.3 V logic level, while the TRIG pin is driven directly from the ESP32 output (acceptable since TRIG is an input to the HC-SR04).

3.6.4 Soil Moisture Sensor Calibration

The capacitive soil moisture sensor (HW-080/HW-103) was empirically calibrated in two extreme physical states to ensure accurate percentage readings from the ESP32's 12-bit ADC:

- Air (0% Moisture): Raw 12-bit ADC value = 3688
- Mineral Water (100% Moisture): Raw ADC value = 1425

The calibration formula programmed into firmware:

$$\text{Moisture (\%)} = ((3688 - \text{analogValue}) / (3688 - 1425)) \times 100.0$$

The ADC1 channel (GPIO 27) was specifically selected for the soil moisture sensor because ADC2 is disabled by the ESP32's internal Wi-Fi driver during network transmission, which would cause erroneous readings on ADC2 channels during data upload.

3.6.5 GPIO Pin Mapping

Component	ESP32 Pin	Logic Level	Power Source	Notes
LM2596 Buck	VIN	5V	7.4V Battery	Main step-down regulator supplying 5V to ESP32
DS18B20 (Temperature)	GPIO 4 (D4)	3.3V	AMS1117	Requires 4.7 kΩ pull-up resistor (R7)
HW-103 Soil AO	GPIO 27 (ADC1)	3.3V	AMS1117	ADC1 to prevent Wi-Fi resource conflict
HW-103 Soil DO	GPIO 26 (D26)	3.3V	AMS1117	Digital threshold alert output
HC-SR04 TRIG	GPIO 5 (D5)	3.3V	LM2596 (5V)	ESP32 triggers ultrasonic burst
HC-SR04 ECHO	GPIO 18 (D18)	5V → 3.3V	LM2596 (5V)	Voltage divider R8/R9/R10 steps signal to 3.3V
Onboard LED	GPIO 2 (D2)	3.3V	Internal	LED ON = Wi-Fi connected; OFF = disconnected

Table 3.5: ESP32 GPIO Pin Mapping and Component Connections

3.6.6 Firmware and Data Transmission

The ESP32 firmware is written in Arduino C++ and implements the following loop:

- Reads all three sensors sequentially every approximately 9 seconds (sum of 500 ms, 2 s, and 2 s delay intervals plus transmission time).

- [illegible]

3.7 Web Application Design

3.7.1 System Architecture

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Backend Server	Python 3 (HTTP Server, requests library)	Local computation server; implements FAO-56 calculations; bridges APIs and frontend
Weather APIs	OpenWeatherMap API + Open-Meteo Archive API	Future forecast (4 days) + Historical rainfall (5 days)
Frontend (Web App)	HTML5, CSS3, Vanilla JavaScript, Chart.js	User interface, live dashboards, 10-day rainfall chart, demand tables

Table 3.6: Web Application System Architecture — Three-Layer IoT Pipeline

3.7.2 Screen-by-Screen Design

The application provides four distinct navigable screens:

- **Screen 1 — Input Configuration:**

User enters or auto-detects GPS coordinates via browser geolocation, selects soil texture type from a dropdown, enters crop type, field area (hectares), channel efficiency, planting date, and the analysis target date. This screen serves as the gateway for all subsequent calculations.

- **Screen 2 — Rainfall Dashboard:**

Displays an interactive 10-day bar chart (5 past days from Open-Meteo Archive + 5 future forecast days from OpenWeatherMap) rendered using Chart.js. Summary metrics include total rainfall, maximum daily rain, and weather classification.

- **Screen 3 — Field Properties:**

Shows calculated soil threshold parameters (Field Capacity, Permanent Wilting Point, and Moisture Threshold) with a visual Soil Moisture Spectrum bar colour-coded into three zones: Wilting Zone, Depletion Zone, and Optimal Zone.

- **Screen 4 — Demand Estimation:**

A tabular 5-day irrigation water requirement forecast showing ET_0 , K_c , ET_c , effective rainfall, and net irrigation demand for each forecast day.

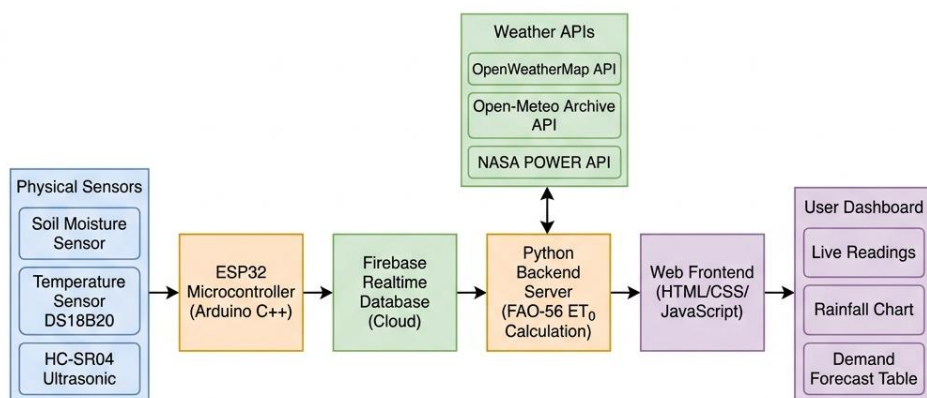
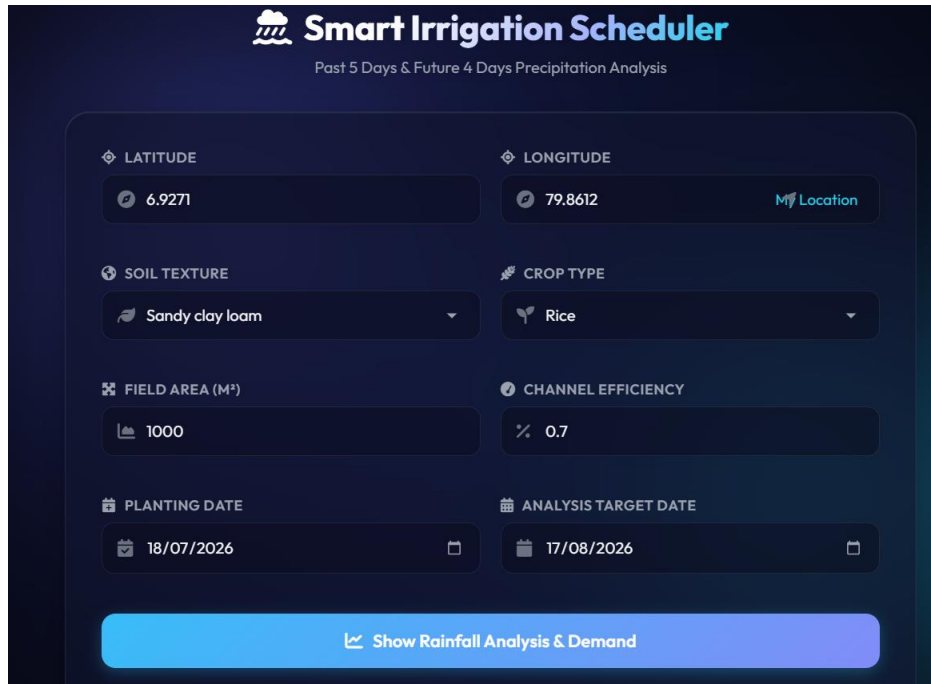


Figure 3.9: Web Application System Architecture — Three-Layer IoT Pipeline



The image shows a web application interface titled "Smart Irrigation Scheduler" with a subtitle "Past 5 Days & Future 4 Days Precipitation Analysis". The interface is designed for configuring irrigation parameters. It features several input fields and dropdown menus arranged in a grid. At the bottom, there is a prominent blue button labeled "Show Rainfall Analysis & Demand".

Parameter	Value
LATITUDE	6.9271
LONGITUDE	79.8612
SOIL TEXTURE	Sandy clay loam
CROP TYPE	Rice
FIELD AREA (M²)	1000
CHANNEL EFFICIENCY	0.7
PLANTING DATE	18/07/2026
ANALYSIS TARGET DATE	17/08/2026

Figure 3.10: Web Application User Interface — Input Configuration Screen



Figure 3.11: Web Application — Rainfall Dashboard (10-Day Chart)

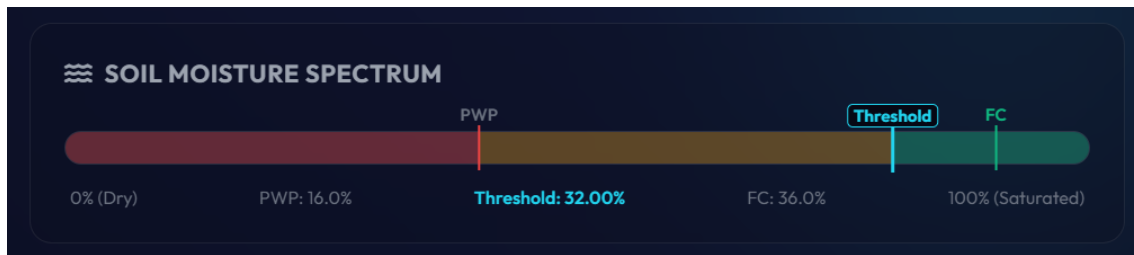


Figure 3.12: Web Application — Field Properties and Soil Moisture Spectrum

Demand Estimation
5-Day Irrigation Water Requirement Forecast

[Rainfall Dashboard](#) [Field Properties](#) [← Modify Inputs](#)

Local JS Engine (Backend Offline) ▲ Elevation: 9 m

Date	Days after Planting	Kc	Crop Water Req. (CWR)	Evaporation (ETO)	Rainfall	Est. Irrigation Demand
2026-08-17	30 days	0.80	0.98 mm/day	1.22 mm/day	5.30 mm	0.0 L/day
2026-08-18	31 days	1.20	2.86 mm/day	2.38 mm/day	2.57 mm	1,144.3 L/day
2026-08-19	32 days	1.20	2.75 mm/day	2.29 mm/day	5.50 mm	0.0 L/day
2026-08-20	33 days	1.20	2.05 mm/day	1.71 mm/day	3.97 mm	0.0 L/day
2026-08-21	34 days	1.20	2.26 mm/day	1.88 mm/day	7.96 mm	0.0 L/day

Figure 3.13: Web Application — Irrigation Demand Estimation Table

4. Results

4.1 IoT Sensor Node Deployment

The IoT sensor node was successfully assembled and deployed in a paddy field environment. The final system consists of the ESP32 microcontroller housed in a weatherproof circuit box mounted on a holding stick above the crop canopy level, with the soil moisture sensor and DS18B20 temperature sensor inserted into the root zone, and the HC-SR04 ultrasonic sensor positioned to measure the distance from a reference point to the water surface within a 5-inch PVC pipe installed in the field.

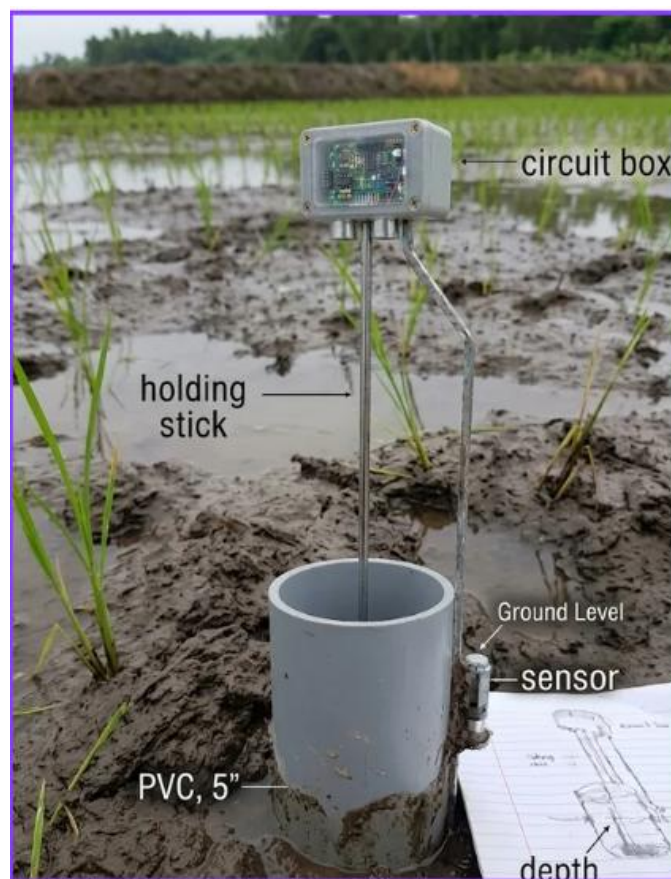


Figure 4.1: Deployed IoT Sensor Node in Paddy Field (circuit box, holding stick, PVC pipe, sensors at ground level)

4.2 Sensor Calibration Results

The capacitive soil moisture sensor was empirically calibrated, recording the following boundary values from the ESP32's 12-bit ADC:

Physical State	Raw 12-bit ADC Reading	Mapped Moisture %
Bone Dry in Air (0% Moisture)	3688	0%
Fully Submerged in Mineral Water (100% Moisture)	1425	100%

Table 4.1: IoT Sensor Calibration Results for Soil Moisture Sensor

The calibration formula was verified to produce accurate and stable percentage readings across the intermediate moisture range during field testing. The HC-SR04 ultrasonic sensor successfully measured water level distances, with the 5 V to 3.3 V voltage divider (R8, R9, R10) operating as designed to protect the ESP32 GPIO.

4.3 Firebase Cloud Telemetry

The ESP32 firmware successfully established Wi-Fi connectivity and transmitted sensor data to the Firebase Realtime Database at the configured URL. The JSON payload (containing moisture percentage, temperature, and water level) was successfully received and stored in the cloud database, as confirmed by the 200 HTTP response codes logged on the serial monitor. The onboard LED (GPIO 2) illuminated correctly to confirm Wi-Fi connection status. The web application frontend retrieved live sensor readings from Firebase and displayed them on the real-time dashboard.

4.4 Irrigation Demand Calculation Results

A sample irrigation water demand forecast was generated using the web application for a representative paddy field configuration: Location — Dry Zone, Sri Lanka; Soil Type — Clay Loam; Crop Type — Paddy Rice; Field Area — 1.5 ha; Canal Efficiency — 70%; Planting Date — Day 30 of the current season.

Day	ET _o (mm/day)	K _c	ET _c (mm/day)	RE (mm/day)	Net Demand (mm/day)
Day 1 (Current)	5.32	0.95	5.05	1.60	4.92
Day 2	5.28	0.95	5.02	0.80	6.03
Day 3	5.45	0.95	5.18	0.00	7.40
Day 4	5.19	0.95	4.93	0.00	7.04
Day 5	5.37	0.95	5.10	2.10	4.29

Table 4.2: Sample Irrigation Demand Calculation Results — 5-Day Forecast

The net irrigation demand was calculated for each forecast day as: $\text{Net Demand} = (\text{ET}_c - \text{RE}) / \text{CE}$, where $\text{CE} = 0.70$. Days with higher rainfall showed significantly reduced

net demand, demonstrating the system's responsiveness to weather-driven changes in irrigation requirements.

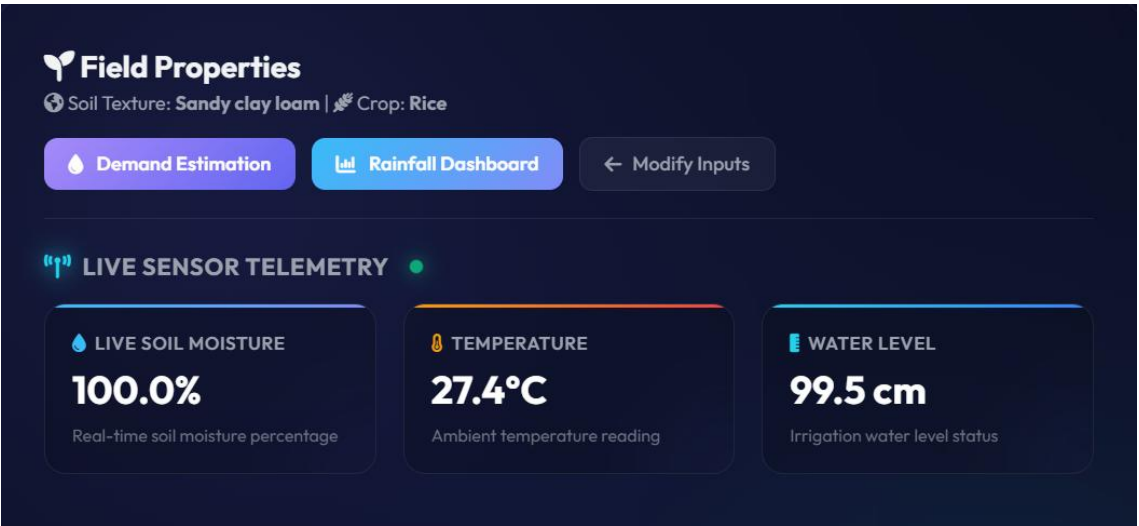


Figure 4.2: Live Sensor Data Received on Web Application Dashboard

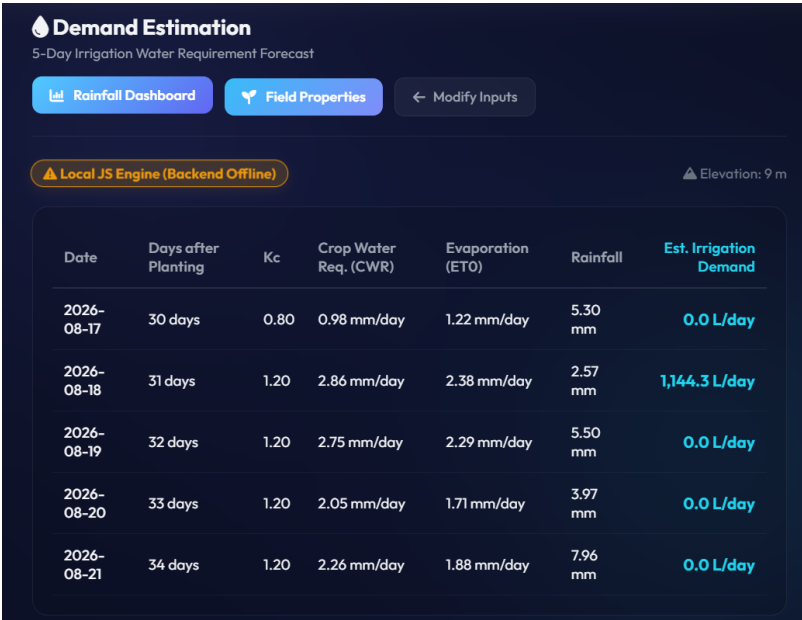


Figure 4.3: Sample Irrigation Demand Forecast Output

4.5 Machine Learning Model Development

As a research component, the LSTM soil moisture prediction model code was developed and trained using publicly available Kaggle soil moisture time-series datasets. The model architecture implemented a two-layer LSTM network with a time window of 60 hourly data points for the shallow soil zone. While the model demonstrated learning from the training data, its validation on local Sri Lankan field conditions is deferred to future work, as discussed in Chapter 7.

5. Discussion

5.1 Demand Calculation Engine

The FAO-56 Penman-Monteith implementation in the Python backend represents a major strength of this project. Unlike simpler temperature-only methods, the Penman-Monteith equation incorporates the full physics of evapotranspiration, radiation, humidity, wind speed, and vapour pressure producing scientifically defensible demand estimates (Allen et al., 1998). The use of the Hargreaves approximation ($kR_s = 0.16$) as a fallback when solar radiation data is unavailable from the API introduces some uncertainty; however, this is a well-documented and accepted practice for agricultural engineering applications (Allen, 1997).

The 70% conveyance efficiency ($CE = 0.70$) applied in the net demand calculation is grounded in established Irrigation Department guidelines for Sri Lanka's dry-zone gravity schemes (Chemjong & Wijesekera, 2017). The effective rainfall coefficient ($RE = 0.67 \times \text{Rainfall}$) is similarly based on the Department of Irrigation's empirical guideline (Wickramaarachchi et al., 2002), ensuring that the demand estimates reflect real-world Sri Lankan conditions rather than generic international standards.

5.2 IoT Hardware Performance

The power system architecture change from a boost converter to a series-battery buck converter topology was a critical engineering decision that resolved the ESP32 brownout reset problem. The LM2596 at 90% efficiency, providing 3 A capacity, offers substantial overhead above the 520 mA peak demand of the ESP32 and ultrasonic sensor combined. The isolation of the analog sensor power supply on a dedicated AMS1117 LDO rail effectively prevented Wi-Fi-induced power fluctuations from corrupting soil moisture and temperature ADC readings.

The voltage divider design for the HC-SR04 ECHO pin is a straightforward but essential safety measure. The calculated output of approximately 3.33 V comfortably falls within the ESP32's 3.3 V logic tolerance, preventing GPIO damage during long-term field operation. The empirical calibration of the soil moisture sensor using physical air and mineral water extremes is more reliable than using manufacturer-provided ADC bounds, which can drift with sensor age and supply voltage variation.

5.3 ML and AHP Frameworks — Limitations and Rationale

The decision to leave the LSTM soil moisture prediction model and the AHP decision framework as research components rather than integrating them into the deployed system is grounded in a realistic appraisal of the constraints of this project. Reliable LSTM model training for local Sri Lankan soil conditions requires continuous seasonal

datasets from deployed sensors across multiple cultivation cycles. The Kaggle datasets used for initial model development represent different soil and climate environments, limiting their transferability.

Similarly, the AHP weighting matrices require calibrated pairwise comparisons from qualified agricultural experts with knowledge of specific local field conditions. While the theoretical framework has been designed and the questionnaire structure has been developed, the expert survey and validation could not be completed within the project timeline. These decisions reflect a responsible approach to ensuring that only validated, scientifically sound components are presented as deployable outputs.

5.4 Comparison with Existing Work

The literature documents several AI-driven irrigation systems, but most are designed for individual field management with continuously connected commercial hardware. This project addresses a unique challenge the collective management of a communal paddy tract (Yaya) which is specific to Sri Lankan irrigation governance. The integration of the AHP mathematical framework for group decision-making is a novel approach that has not been widely implemented in the Sri Lankan context. The SCMARL framework explored in this project aligns with cutting-edge research (Agyeman et al., 2024), demonstrating awareness of the state of the art in precision irrigation.

The web application architecture combining a local Python computation server with Firebase cloud storage, dual API data fusion (historical + forecast), and a SPA frontend, represents a pragmatic and deployable solution for the resource constraints of small-scale Sri Lankan agricultural settings, where dedicated cloud-based servers may not be economically viable.

6. Conclusions

This project has successfully designed, developed, and deployed a comprehensive proof-of-concept AI-driven water management system for paddy agriculture in Sri Lanka. The following key conclusions are drawn from the work completed:

- The FAO-56 Penman-Monteith demand calculation engine, implemented in Python and integrated with OpenWeatherMap and Open-Meteo weather APIs, produces scientifically grounded 5-day irrigation demand forecasts adapted to Sri Lankan dry-zone conditions. The engine correctly applies local empirical parameters ($RE = 0.67$, $CE = 0.70$) from the Department of Irrigation.
- The soil moisture threshold calculation module correctly computes the irrigation trigger point ($\theta_{\text{trigger}} = FC - [MAD \times (FC - PWP)]$) for paddy rice ($MAD = 0.20$), providing farmers with a precise boundary for irrigation initiation based on their field's soil texture.
- The research into LSTM-based soil moisture prediction identified the -40 cm depth cutoff as a critical modelling boundary and established an LSTM architecture suitable for future deployment when local field-calibrated datasets become available.
- The AHP multi-criteria decision framework provides a mathematically rigorous approach to aggregating the heterogeneous irrigation needs of multiple fields within a communal Yaya into a single collective scheduling decision. The stage-wise parameter impact matrix demonstrates that parameter importance varies significantly across the four paddy growth stages, requiring dynamic weighting.
- The ESP32 IoT sensor node was successfully designed with a stable power architecture (LM2596 + AMS1117), empirically calibrated sensors, and reliable Firebase Realtime Database cloud telemetry, providing real-time field data to the web application.
- The full-stack web application successfully integrates all computational components with live IoT telemetry and a responsive user interface, demonstrating a complete end-to-end IoT data pipeline from physical sensors in a paddy field to a web-based decision-support dashboard.

Overall, this project establishes a multi-disciplinary engineering foundation for AI-driven precision irrigation management in Sri Lanka, addressing the unique structural, operational, and agronomic characteristics of the country's paddy irrigation systems. It demonstrates that a data-driven approach combining civil hydraulic engineering, agricultural science, machine learning, IoT hardware, and web development is both technically feasible and practically relevant for improving water use efficiency in Sri Lankan agriculture.

7. Future Work

The current system represents a proof-of-concept implementation. Several technical and methodological enhancements are planned to transition it into an industrial, field-deployable precision agriculture solution.

7.1 Hardware Optimisation and Long-Range Telemetry

- **LoRaWAN Communication:** Replace the current Wi-Fi telemetry with LoRa / LoRaWAN transceivers, enabling ultra-long-range data transmission (up to 5–10 km in open agricultural terrain) without relying on cellular subscriptions or sparse farm Wi-Fi coverage.
- **Power Management and Duty Cycling:** Implement low-power firmware routines utilising ESP32 deep-sleep modes, with sensors sampled periodically (every 3–6 hours) to maximise battery life and enable solar-harvesting autonomy.
- **Custom PCB Design:** Consolidate discrete prototyping modules into a dedicated, miniaturised custom Printed Circuit Board (PCB) with IP67-rated ingress protection for resilience against agricultural field conditions.

7.2 Real-World Machine Learning Scaling and Integration

- **Empirical Validation with In-Situ Field Data:** Transition LSTM model evaluation from static/synthetic repositories to continuous datasets collected directly across full seasonal cultivation cycles in local Sri Lankan paddy environments.
- **Dynamic Parameter Expansion:** Broaden the predictive feature space to include dynamic Leaf Area Index (LAI), NDVI time-series from remote sensing, electrical conductivity (EC), and localised solar radiation profiles.
- **Multi-Step Horizon Tuning:** Optimise LSTM look-back windows and recursive prediction horizons to reliably forecast root-zone moisture depletion 3–7 days ahead under extreme monsoon and dry-spell transitions.

7.3 Multi-Criteria Decision Validation and AHP Integration

- **Empirical AHP Calibration:** Conduct rigorous validation of the AHP weighting matrices against agronomic expert feedback from the Department of Agriculture and Department of Irrigation, and against historical yield records from completed cultivation seasons.
- **Dynamic Decision Thresholds:** Integrate the AHP priority scoring directly into the closed-loop control system, dynamically balancing trade-offs between water conservation quotas, pumping energy costs, and crop stress thresholds in real time.
- **Automated Expert Survey System:** Develop a digital interface for collecting and aggregating expert pairwise comparison responses, automating the CR validation and weight synthesis process.

7.4 Zonal Scalability and Multi-Field Management

- **Distributed Multi-Field Topology:** Expand the single-node, single-field architecture into a multi-node mesh topology capable of managing multiple heterogeneous plots simultaneously, mapping to the Yaya-level group decision model.
- **Zonal Variable Rate Irrigation (VRI):** Enable the central scheduler to generate zone-specific irrigation commands tailored to varying soil textures, crop varieties, and growth stages across different paddy plots within a single Yaya, from a unified web interface.
- **Integration with Reservoir Management:** Link the field-level demand forecasts with reservoir capacity data to provide system-wide, season-long water allocation planning for the full irrigation scheme.

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Appendices

Appendix A: ESP32 Firmware Code

The complete firmware code for the ESP32 IoT sensor node (setup2_data_send.ino) is presented below. This code implements Wi-Fi connectivity, sensor reading routines for the HC-SR04 ultrasonic sensor, DS18B20 temperature sensor, and HW-103 capacitive soil moisture sensor, and transmits sensor data to Firebase Realtime Database via HTTPS.

```
// ESP32 IoT Sensor Node Firmware
// Libraries
#include <WiFi.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>
#include <OneWire.h>
#include <DallasTemperature.h>

// Configuration
const char* ssid      = "YOUR_WIFI_SSID";
const char* password  = "YOUR_WIFI_PASSWORD";
const char* firebaseUrl =
    "https://smart-irrigation-schedul-d2add-default-rtdb.firebaseio.com/sensor-
latest.json";

// Pin Definitions
const int TRIG_PIN      = 5;    // HC-SR04 Trigger
const int ECHO_PIN      = 18;   // HC-SR04 Echo (with voltage divider)
#define ONE_WIRE_BUS    4       // DS18B20 Data
const int ANALOG_PIN    = 27;   // Soil Moisture AO (ADC1)
const int DIGITAL_PIN   = 26;   // Soil Moisture DO
const int ONBOARD_LED_PIN = 2;  // Status LED

// Calibration constants
const int dryValue = 3688;  // Air (0% moisture)
const int wetValue = 1425;  // Mineral water (100% moisture)

// Sound speed
#define SOUND_SPEED 0.034

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

long duration;
float distanceCm;

void blank_trigger() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);
}

void setup() {
    Serial.begin(115200);
```

```

pinMode(ONBOARD_LED_PIN, OUTPUT);
digitalWrite(ONBOARD_LED_PIN, LOW);
WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) { delay(500); }
digitalWrite(ONBOARD_LED_PIN, HIGH);
pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
pinMode(DIGITAL_PIN, INPUT);
sensors.begin();
}

void loop() {
  // Ultrasonic water level measurement
  blank_trigger();
  duration = pulseIn(ECHO_PIN, HIGH);
  distanceCm = duration * SOUND_SPEED / 2;
  delay(500);

  // Temperature reading
  sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);
  delay(2000);

  // Soil moisture reading and calibration
  int analogValue = analogRead(ANALOG_PIN);
  float moisturePercentage = (float)((dryValue - analogValue) /
                                     (float)(dryValue - wetValue)) * 100.0;
  delay(2000);

  // Firebase data transmission
  if (WiFi.status() == WL_CONNECTED) {
    WiFiClientSecure client;
    client.setInsecure();
    HTTPClient http;
    http.setTimeout(5000);
    http.begin(client, firebaseUrl);
    http.addHeader("Content-Type", "application/json");
    String jsonPayload = "{"moisture":" + String(moisturePercentage) +
                          ", "temperature":" + String(tempC) +
                          ", "waterlevel":" + String(distanceCm) + "}";
    int httpResponseCode = http.PUT(jsonPayload);
    http.end();
    digitalWrite(ONBOARD_LED_PIN, HIGH);
  }
  delay(5000);
}

```

Appendix B: Python Demand Calculator Code

The core irrigation demand calculation function implemented in `demand_calculator.py` implements the FAO-56 Penman-Monteith ETo calculation, crop coefficient lookup, effective rainfall computation, and net irrigation demand estimation.

Appendix C: AHP Pairwise Comparison Questionnaire

The following questionnaire was designed for expert elicitation of irrigation parameter importance weights for the AHP decision model.

Set A: Crop Type Effect (Level 2 Weights)

Question: 'Considering the crop type in Field n, compare the relative importance of these parameters in determining overall irrigation need.' Rate on Saaty's 1–9 scale where: 1 = equally important, 3 = moderately more important, 5 = strongly more important, 7 = very strongly more important, 9 = extremely more important.

Set B: Plant Stage Sensitivity (Level 3 Weights)

Question: 'Considering the current plant growth stage, compare which parameter is more critical in driving a YES decision to trigger irrigation.' Rate on Saaty's 1–9 scale.

Appendix D: Web Application Technology Stack

Category	Technology
Frontend	HTML5, CSS3, Vanilla JavaScript
Charting Library	Chart.js (via CDN)
Icons	Font Awesome 6
Backend Server	Python 3 (HTTP Server, requests library)
Cloud Database	Google Firebase Realtime Database
Weather APIs	OpenWeatherMap API, Open-Meteo Archive API
Solar Radiation API	NASA POWER API
IoT Embedded Platform	Arduino C++ (ESP32, WiFi, HTTPClient, DallasTemperature, OneWire)
Deployment	Firebase Hosting